Noise Traders Are Not a Primitive

Aggregation Space, Risk Preferences, and the Revelation of Information

Matthijs Breugem

Nyenrode Business University

April 27, 2026

Abstract

A long tradition in financial economics treats noise traders as a structural prerequisite for partial revelation of information through prices: without exogenous shocks to supply or demand, prices fully aggregate private signals and the Grossman–Stiglitz paradox bites. This paper shows that the prerequisite is preference-specific. Full revelation in the canonical rational-expectations model is a knife-edge property of constant absolute risk aversion (CARA), driven by the alignment between CARA market clearing and the Bayesian sufficient statistic in log-odds space. Under any non-CARA preference in the constant relative risk aversion (CRRA) class, market clearing aggregates in probability space rather than log-odds, the Jensen gap between these two spaces becomes the source of revelation failure, and equilibrium is generically partially revealing without any noise. I formalise the mechanism analytically in a binary-state model with $K \geq 3$ informed groups, prove that CARA is the unique full-revelation preference class within CRRA, and verify numerically that partial revelation survives full rational-expectations learning via a contour-integration solution of the fixed point. The implications are immediate: information has positive value, trade volume is positive, and the Grossman–Stiglitz paradox is resolved within the rational class without recourse to liquidity or behavioural noise.

Keywords: rational expectations equilibrium, partial revelation, noise traders, CARA, CRRA, Grossman–Stiglitz paradox, aggregation space.
JEL codes: D82, D83, G14.

1 Introduction

Half a century after Grossman and Stiglitz (1980), the textbook resolution of the informational paradox in asset markets remains the same: prices cannot aggregate private signals perfectly because noise — exogenous shocks to supply, liquidity demand, or trader population — masks the information content of order flow. From Hellwig (1980) and Diamond and Verrecchia (1981) through Kyle (1985) and the modern microstructure literature, the conditional partial revelation of information is engineered by an unmodelled population whose trades are unrelated to fundamentals. The modelling convention is so entrenched that introductory treatments routinely list "noise traders"

alongside risk aversion and information asymmetry as a *primitive* of the rational-expectations framework.

This paper makes a simple but consequential point: that primitive is not necessary. In a fully rational economy with no liquidity shocks, no behavioural traders, and zero net supply, partial revelation arises endogenously from preferences alone. What is required is that agents do not have constant absolute risk aversion. Once preferences exit the CARA class, the mapping from private information to equilibrium prices fails to identify the Bayesian sufficient statistic, and prices reveal partial information for purely aggregational reasons.

The mechanism is best stated in terms of an *aggregation space*. Under any preference class, equilibrium prices aggregate private posteriors via market clearing. Under CARA, demand is linear in the log-odds of the posterior, and clearing therefore averages log-odds. The Bayesian sufficient statistic in the canonical Gaussian-signal model is itself a sum of log-likelihood ratios. The two spaces coincide — log-odds in, log-odds out — and prices fully reveal that statistic. Under any other CRRA preference, demand is nonlinear in log-odds and approximately linear in probability; clearing therefore averages in probability space. The probability average of a logistic transform is not the logistic transform of the average,

$$\frac{1}{K}\sum_{k=1}^{K}\Lambda(\tau u_k) \neq \Lambda\left(\frac{1}{K}\sum_{k=1}^{K}\tau u_k\right),$$

and the Jensen gap on the right is the size of the revelation failure. CARA is the boundary case in which the gap closes because the link from posteriors to demand is itself the inverse logistic — a coincidence of functional form, not an economic primitive.

The contributions of the paper are four. First, I show analytically that in a binary-state model with $K \geq 3$ groups, the CARA market-clearing condition aggregates private posteriors in log-odds for arbitrary heterogeneity in risk aversion, signal precision, and wealth, and is fully revealing of the Bayesian sufficient statistic when risk aversion is homogeneous (Proposition 1). Heterogeneous risk aversion introduces a separate route to partial revelation, but the underlying log-odds aggregation is robust and continues to identify CARA as the boundary case. Second, I show that under any CRRA preference with $\gamma \neq \infty$ the no-learning equilibrium is partially revealing, and I derive the third-order asymptotic expansion of the Jensen gap (Propositions 2, 3). Third, I solve the full rational-expectations equilibrium numerically using a contour-integration fixed point that identifies the unknown price function $P[i, j, l]$ on a three-dimensional grid; the fixed point converges, and partial revelation survives at the equilibrium that is consistent with rational learning from prices (Proposition 5). Fourth, I trace the welfare consequences and the resolution of the Grossman–Stiglitz paradox: under CRRA there is positive trade volume, the value of private information is strictly positive and increasing in precision, and the paradox dissolves without recourse to noise (Propositions 6–8).

Relation to the literature. The closest precursor is DeMarzo and Skiadas (1998), who establish that in the Gaussian-CARA model the rational-expectations equilibrium is unique and fully revealing, and Breon-Drish (2015), who extend full revelation to the exponential family of signal distributions with CARA demands. Their results are usually read as evidence that revelation is generic and that the role of noise traders is to introduce frictions strong enough to sustain non-trivial informational rents. The present paper takes the opposite view: full revelation is a knife-edge property of CARA combined with appropriate signal distributions, and the broader rational class delivers partial revelation without any frictions. Vives (2011) and the strategic-information literature offer one route around CARA via private values. Recent work by Albagli et al. (2024) characterises non-Gaussian equilibria with CARA-like demands; the mechanism here is complementary, operating on the preference rather than the distributional dimension. Numerically, the contour-integration fixed point I introduce builds on the projection method of Breugem and Buss (2019) for non-standard rational-expectations problems.

Organisation. Section 2 sets out the model and the demand characterisations. Section 3 proves the no-learning benchmark results and presents the smooth-transition table that maps the move from CARA to log utility. Section 4 develops the contour method, states the rational-expectations result, and reports converged posteriors. Section 5 treats welfare, the value of information, and the Grossman–Stiglitz paradox. Section 6 dissects the three mechanisms that drive partial revelation under heterogeneity. Section 7 relates the results to the wider literature, and Section 8 concludes. All proofs are in Appendix A; the numerical implementation of the contour fixed point is in Appendix B.

2 Model

A binary state of the world $v \in \{0, 1\}$ is realised at $t = 1$. At $t = 0$ a single risky asset trades against a riskless numeraire; the asset pays v units of the numeraire at $t = 1$ and is in zero net supply, $\bar{z} = 0$. There are no noise traders, no liquidity shocks, and no exogenous supply variation: the entire equilibrium is determined by the preferences and signals of the rational agents.

2.1 Information structure

There are $K \geq 3$ groups of informed agents indexed by $k = 1, \ldots, K$. Each group k has continuum measure normalised to one within the group and observes a private signal

$$s_k \;=\; v + \varepsilon_k, \qquad \varepsilon_k \sim \mathcal{N}(0, 1/\tau_k), \tag{1}$$

where ε_k is independent across k and independent of v. The prior is uniform, $\mathbb{P}(v = 1) = \frac{1}{2}$. It will be convenient to work with the centred signal $u_k \equiv s_k - \frac{1}{2}$, which satisfies $u_k \mid v = 1 \sim \mathcal{N}(\frac{1}{2}, 1/\tau_k)$

and $u_k \mid v = 0 \sim \mathcal{N}(-\frac{1}{2}, 1/\tau_k)$. The signal density is

$$f_v(u_k) \;=\; \sqrt{\tfrac{\tau_k}{2\pi}}\,\exp\!\Big(-\tfrac{\tau_k}{2}\big(u_k - v + \tfrac{1}{2}\big)^2\Big)\,, \tag{2}$$

and the private log-likelihood ratio is $\ln[f_1(u_k)/f_0(u_k)] = \tau_k u_k$.

By Bayes' rule, the private posterior is

$$\mu_k \;\equiv\; \mathbb{P}(v = 1 \mid u_k) \;=\; \Lambda(\tau_k u_k), \tag{3}$$

where $\Lambda(z) = 1/(1 + e^{-z})$ is the logistic function. The Bayesian sufficient statistic for the joint likelihood ratio across all K signals is

$$T^\star \;\equiv\; \sum_{k=1}^{K} \tau_k u_k, \tag{4}$$

since $\ln \mathbb{P}(v = 1 \mid u_1, \ldots, u_K)/\mathbb{P}(v = 0 \mid u_1, \ldots, u_K) = T^\star$. Throughout, $\mathrm{logit}(p) \equiv \ln[p/(1-p)]$ denotes the inverse logistic.

2.2 Preferences and demands

Each agent in group k has initial wealth $W_k > 0$ and maximises expected utility over terminal wealth

$$W_k - x_k p + x_k v, \tag{5}$$

where x_k is her holding of the risky asset and $p \in (0, 1)$ is the equilibrium price. Two preference families are central. The CRRA family parametrised by $\gamma \in (0, \infty)$ is

$$U_{\mathrm{CRRA}}(W) \;=\; \frac{W^{1-\gamma}}{1-\gamma}, \tag{6}$$

with the logarithmic limit $U(W) = \ln W$ at $\gamma = 1$. The CARA family parametrised by $\alpha > 0$ is

$$U_{\mathrm{CARA}}(W) \;=\; -\exp(-\alpha W). \tag{7}$$

We will refer informally to "CARA" as the $\gamma \to \infty$ limit of CRRA; this is justified because the binary structure of the asset implies that optimal demands depend on wealth only through a CARA-like log-odds quotient in that limit.

Demand functions. Pointwise maximisation of expected utility yields closed-form demands.

Lemma 1 (CARA demand)**.** *Under preferences* (7)*, the optimal holding of an agent with posterior* μ_k *at price* p *is*

$$x_k^{\mathrm{CARA}}(\mu_k, p) \;=\; \frac{1}{\alpha}\big[\mathrm{logit}(\mu_k) - \mathrm{logit}(p)\big]. \tag{8}$$

Lemma 2 (CRRA demand)**.** *Under preferences* (6) *with* $\gamma \in (0, \infty)$*, the optimal holding of an agent with wealth* W_k*, posterior* μ_k*, and price* p *is*

$$x_k^{\mathrm{CRRA}}(\mu_k, p) \;=\; \frac{W_k\,(R_k - 1)}{(1-p) + R_k\,p}, \qquad R_k \;\equiv\; \exp\!\left(\frac{\mathrm{logit}(\mu_k) - \mathrm{logit}(p)}{\gamma}\right). \tag{9}$$

At $\gamma = 1$ *this reduces to the log-utility demand* $x_k = W_k(\mu_k - p)/[p(1-p)]$*.*

The decisive feature of (8) is that the CARA demand is linear in the difference $\mathrm{logit}(\mu_k) - \mathrm{logit}(p)$. The CRRA demand is *nonlinear* in this difference; in the small-bet limit it is approximately linear in the probability gap $\mu_k - p$, but the small-bet approximation is local and the global market-clearing condition cannot be reduced to a log-odds aggregator.

2.3 Equilibrium

A no-learning equilibrium of the model is a price p that clears the market when each group k uses only its own private posterior μ_k:

$$\sum_{k=1}^{K} x_k(\mu_k, p) \;=\; 0. \tag{10}$$

A rational-expectations equilibrium (REE) is a price function $p(u_1, \ldots, u_K)$ such that each group k uses the conditional posterior

$$\mu_k^{\mathrm{REE}} \;\equiv\; \mathbb{P}(v = 1 \mid u_k,\, p(u_1, \ldots, u_K)) \tag{11}$$

in place of μ_k and the market clears at every realisation. Section 4 develops the contour fixed-point characterisation of this equilibrium. Throughout, we focus on equilibria in which p depends on the signal vector only through prices the agents can observe (a standard internal consistency requirement).

2.4 The aggregation space

The unifying observation of this paper is that no-learning equilibrium prices aggregate private posteriors in a space determined by the linearisation of demand. To make this precise, define the *aggregation map* $\Phi : \mathbb{R}^K \to \mathbb{R}$ implicitly by the market-clearing condition: Φ is the function such that $\mathrm{logit} p = \Phi(\mathrm{logit}\mu_1, \ldots, \mathrm{logit}\mu_K)$ or, equivalently, $p = \Psi(\mu_1, \ldots, \mu_K)$. Under CARA, (8) and (10) imply

$$\mathrm{logit} p \;=\; \sum_{k=1}^{K} w_k\, \mathrm{logit}\mu_k, \qquad w_k \;\propto\; 1/\alpha_k, \tag{12}$$

which is a weighted average in log-odds space. Under log utility, (9) with $\gamma = 1$ implies

$$p \;=\; \sum_{k=1}^{K} w_k\, \mu_k, \qquad w_k \;\propto\; W_k, \tag{13}$$

which is a weighted average in probability space. The Bayesian sufficient statistic (4) *is* a sum in log-odds space. The CARA aggregator therefore matches the sufficient statistic exactly; the log-utility aggregator does not, and the gap is informational.

3 The No-Learning Benchmark

This section establishes the analytical core of the paper in the simpler no-learning environment, where each agent uses only her own private signal. The conclusions extend to the full REE in Section 4.

3.1 CARA implies full revelation

Proposition 1 (CARA log-odds aggregation and full revelation)**.** *Fix any $K \geq 1$ and any parameter vector $(\alpha_k, \tau_k, W_k)_{k=1}^K$. The unique no-learning equilibrium under CARA preferences aggregates private posteriors in log-odds:*

$$\text{logit}p \;=\; \sum_{k=1}^{K} w_k \,\text{logit}\mu_k \;=\; \sum_{k=1}^{K} w_k\, \tau_k\, u_k, \qquad w_k \;=\; \frac{\alpha_k^{-1}}{\sum_j \alpha_j^{-1}}. \tag{14}$$

When risk aversion is homogeneous across groups, $w_k = 1/K$ and $\text{logit}p = T^\star/K$ is a known scalar multiple of the Bayesian sufficient statistic $T^\star$ of (4). The equilibrium is then fully revealing in the sense that $\mathbb{P}(v = 1 \mid p) = \mathbb{P}(v = 1 \mid u_1, \ldots, u_K)$ for almost every realisation. With heterogeneous α_k the no-learning price reveals a different linear combination of the signals — a fourth route to partial revelation, discussed in Section 6 — but, because the candidate fully-revealing price function $p = \Lambda(T^\star)$ satisfies market clearing identically (zero trade) regardless of (α_k), the rational-expectations *equilibrium remains fully revealing for arbitrary heterogeneous CARA parameters.*

The proof, in Appendix A, exploits the linearity of (8) to reduce (10) to a linear equation in $\text{logit}p$, then composes with (3) to obtain the weighted log-odds form. Under homogeneous risk aversion, full revelation follows because $T^\star$ is a sufficient statistic for the joint posterior $\mathbb{P}(v = 1 \mid u_1, \ldots, u_K)$ and any affine function of $T^\star$ is informationally equivalent. The log-odds aggregation form generalises DeMarzo and Skiadas (1998) from the Gaussian-asset CARA model to a binary asset, and a parallel extension obtains for any signal distribution in the exponential family with CARA preferences (Breon-Drish, 2015); the proof in Appendix A sketches this extension.

3.2 Non-CARA implies partial revelation

Proposition 2 (Non-CARA partial revelation)**.** *Fix $K \geq 3$, $\gamma = 1$, and homogeneous (τ, W). The no-learning equilibrium under log utility satisfies*

$$p \;=\; \frac{1}{K} \sum_{k=1}^{K} \Lambda(\tau u_k). \tag{15}$$

The equilibrium price p is not a function of $T^\star = \sum \tau u_k$ alone: there exist signal realisations with the same $T^\star$ but different p. Hence the equilibrium is partially revealing.

The price aggregates posteriors in probability space, while the sufficient statistic lives in log-odds space. The two coincide if and only if Λ acts linearly on the relevant summation, which by Jensen's inequality fails strictly whenever Λ is nonlinear — that is, whenever τu_k has positive variance.

3.3 The Jensen gap

The proximity to full revelation under CRRA can be quantified by an asymptotic expansion of the gap between the actual price and the fully-revealing benchmark.

Proposition 3 (Jensen-gap expansion)**.** *Under the conditions of Proposition 2, the deviation of the no-learning price from the fully-revealing CARA price $p^{\mathrm{CARA}} = \Lambda(T^\star/K)$ admits the third-order expansion*

$$p - p^{\mathrm{CARA}} \;=\; -\frac{\tau^3}{48\,K}\left(U_3 - \frac{U_1^3}{K^2}\right) \;+\; O(\tau^5), \tag{16}$$

where $U_n \equiv \sum_{k=1}^{K} u_k^n$. The leading-order term is strictly nonzero whenever the empirical distribution of $u_1, \ldots, u_K$ has nonzero excess third central moment.

The expansion makes precise the informational content of the gap: the ordering of u_k, not just their sum, enters the equilibrium price. The agent observing p alone cannot disentangle a low-$T^\star$ realisation with a skewed signal vector from a high-$T^\star$ realisation with a symmetric one. This is the source of revelation failure under any non-CARA preference.

3.4 Smooth transition from CARA to log

Propositions 1–3 together imply that the revelation deficit, measured by $1 - R^2$ in the regression of $T^\star$ on p, is continuous and monotone in the CRRA parameter γ.

Proposition 4 (Smooth transition)**.** *Fix (K, τ, W) and parametrise preferences by $\gamma \in (0, \infty]$, with $\gamma = \infty$ denoting CARA. The revelation deficit $1 - R^2\big(T^\star\,;\,p_\gamma\big)$ is continuous in γ, strictly decreasing on $(0, \infty)$, and equal to zero at $\gamma = \infty$. CARA is therefore a knife-edge case within the rational class.*

Figure 1 reports the deficit as a function of the signal precision τ for three levels of γ together with the CARA benchmark. The deficit is strictly positive for every finite γ and collapses smoothly to zero as $\gamma \to \infty$.

Figure 2 provides a complementary view as a heatmap over the (γ, τ) plane: the locus $1 - R^2 = 0$ is exactly the right edge $\gamma = \infty$, and the deficit grows continuously as one moves into the interior.

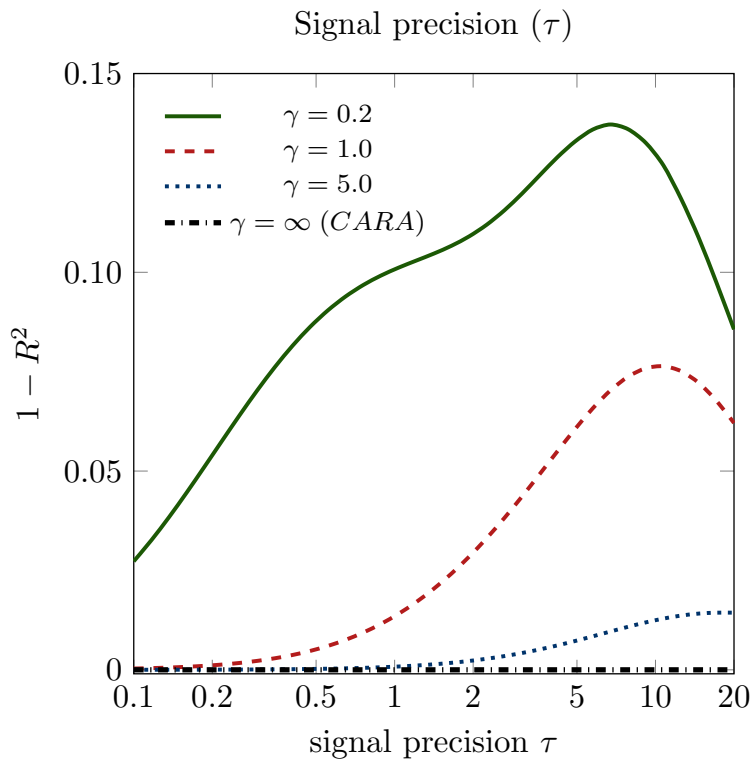


Figure 1: Revelation deficit $1 - R^2$ in the no-learning equilibrium as a function of signal precision τ, for $\gamma \in \{0.2, 1, 5\}$ and the CARA benchmark ($\gamma = \infty$). $K = 3$, $W = 1$, signals on a grid of $G = 20$. CARA produces zero deficit for every τ; every finite γ produces strictly positive deficit, monotone in γ.

Table 1: No-learning revelation deficit $1 - R^2$ across γ and τ. Computed exactly on a $G = 20$ grid. CARA ($\gamma = \infty$) is the knife-edge.

γ	$\tau = 0.5$	$\tau = 1.0$	$\tau = 2.0$
0.1	0.146	0.145	0.137
0.3	0.044	0.070	0.090
0.5	0.016	0.038	0.062
1.0	0.004	0.013	0.029
3.0	0.000	0.002	0.006
10.0	0.000	0.000	0.001
∞ (CARA)	0.000	0.000	0.000

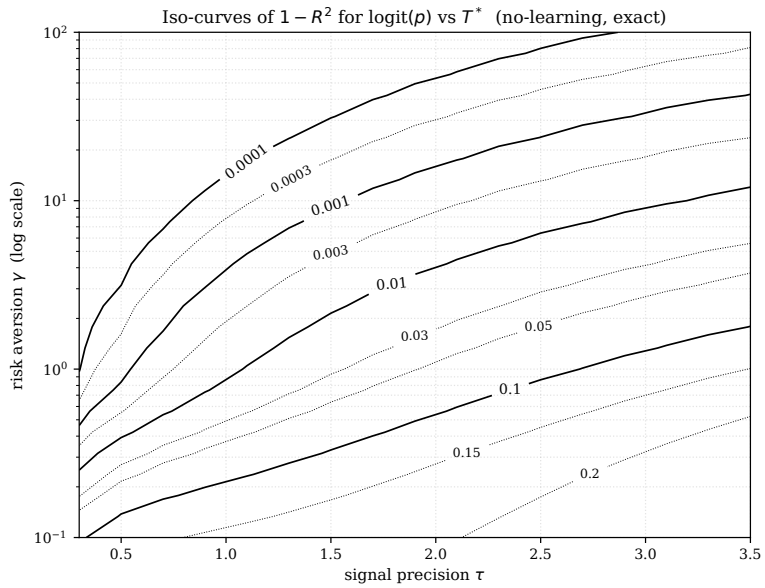


Figure 2: Revelation deficit on the (γ, τ) plane. The CARA line sits at the right edge; the deficit grows monotonically as γ falls.

3.5 Robustness

The CARA full-revelation result extends, with weighted aggregation, to heterogeneous parameters. The non-CARA partial-revelation result extends, in the natural direction, to any preference outside CARA in the CRRA class and to a wide range of signal distributions. Two robustness exercises confirm this.

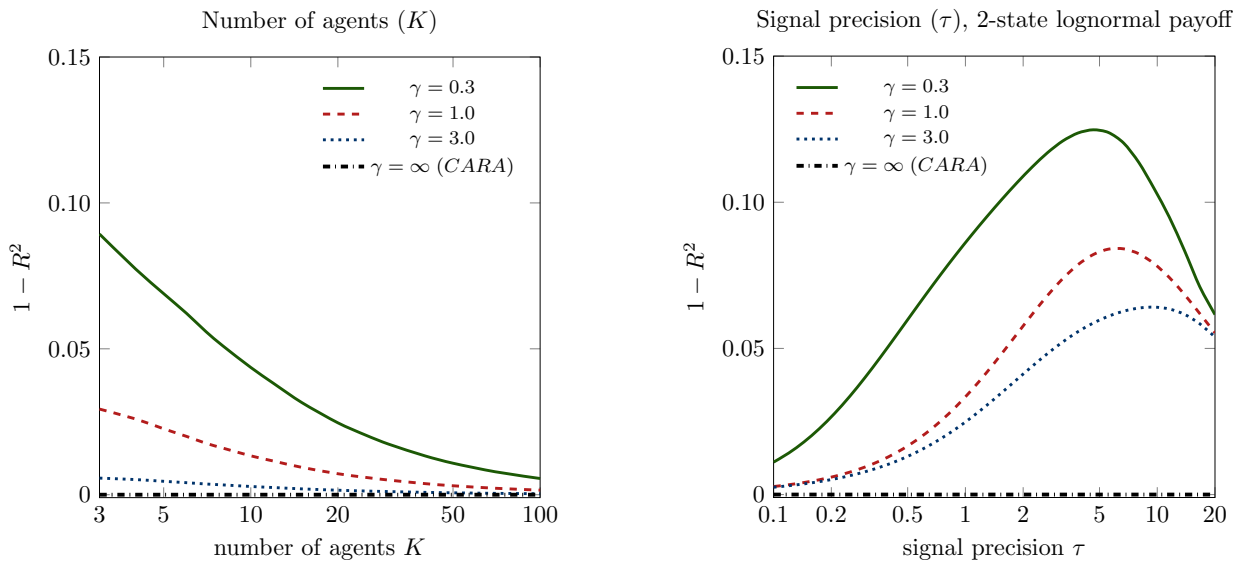
4 Full Rational-Expectations Equilibrium

The no-learning benchmark establishes the analytical core. The next step is to verify that partial revelation survives once agents update on the price itself, that is, in the full REE defined by (11). This section formulates the REE as a fixed point on the price function $P[i, j, l]$, develops a contour-integration solution method, and reports converged equilibria.

4.1 The price-function fixed point

For computational purposes, fix $K = 3$ and discretise the centred-signal support uniformly at G grid points $\{u_i\}_{i=1}^{G}$. The unknown is the price array $P \in [0, 1]^{G^3}$, where $P[i, j, l]$ is the equilibrium price at the realisation (u_i, u_j, u_l).

Each agent observes her own signal and the price. Conditioning on the price, agent k at signal u_{i_k} infers the set of other-signal pairs that are consistent with the observed price. This set is the level set of P at the price level, restricted to the appropriate two-dimensional slice. We call this set the *contour* traced by the agent.



(a) Number of agent groups K.

(b) Lognormal payoff variant.

Figure 3: Robustness of the knife-edge. Panel (a): the deficit declines roughly as $1/K$ but never vanishes for any finite γ; CARA is zero for every K. Panel (b): under a lognormal payoff, the same pattern obtains, indicating the result is not specific to the binary parametrisation.

The contour fixed point is the requirement that, when each agent integrates the prior signal density along her contour to form a conditional posterior and then submits her optimal demand, the market clears at every realisation (i, j, l) at the conjectured price $P[i, j, l]$. Symbolically:

$$P \;=\; \Phi(P), \qquad \Phi(P)[i,j,l] \;=\; \big\{\, p \,:\, \textstyle\sum_{k=1}^{3} x_k(\mu_k^P(p), p) = 0 \,\big\}, \tag{17}$$

where $\mu_k^P(p)$ is the conditional posterior of agent k obtained by integrating along her contour, given the conjectured price function P and the observed price p.

4.2 The contour integration

Agent 1 at her own signal u_i extracts the slice $P[i, \cdot, \cdot]$, which is a $G \times G$ matrix of prices in (u_j, u_l) space. The set $\big\{(u_j, u_l) \,:\, P[i, j, l] = p\big\}$ is a one-dimensional curve. We trace it numerically by a two-pass algorithm:

1. Pass A. For each grid point $u_{j'}$, treat the row $P[i, j', \cdot]$ as a function of u_l and root-find $u_l^c(j')$ such that $P[i, j', u_l^c] = p$.

2. Pass B. For each grid point $u_{l'}$, treat the column $P[i, \cdot, l']$ as a function of u_j and root-find $u_j^c(l')$.

Each crossing contributes the joint signal density $f_v(u_j^c)\, f_v(u_l^c)$ to a state-conditional accumulator $A_v^{(1)}$. Averaging the two passes,

$$A_v^{(1)}(i,p) \;=\; \tfrac{1}{2}\left[\sum_{j'} f_v(u_{j'})\, f_v(u_l^c(j')) + \sum_{l'} f_v(u_j^c(l'))\, f_v(u_{l'})\right]. \tag{18}$$

Agent 1's posterior is then

$$\mu_1(i,p) \;=\; \frac{f_1(u_i)\, A_1^{(1)}(i,p)}{f_0(u_i)\, A_0^{(1)}(i,p) + f_1(u_i)\, A_1^{(1)}(i,p)}. \tag{19}$$

Agents 2 and 3 proceed analogously on their own slices, $P[\cdot, j, \cdot]$ and $P[\cdot, \cdot, l]$ respectively. The market-clearing condition $\sum_k x_k(\mu_k(\cdot, p), p) = 0$ then determines a new price at every (i, j, l), completing one application of Φ.

A central observation is that the integrals (18) employ *prior* densities; all conditioning on the price is encoded in the contour itself, that is, in the price function P. The unknown of the problem is therefore P and only P; there is no separate fixed point in posteriors.

4.3 Why CARA gives a straight contour and CRRA gives a curved one

Under CARA, by Proposition 1, $\mathrm{logit} P[i, j, l] = (1/3)(\tau u_i + \tau u_j + \tau u_l)$, so the level sets of P in the (u_j, u_l) slice are the lines $\tau u_j + \tau u_l = c$. Every point on such a contour has the same Bayesian sufficient statistic $T^\star$. The density ratio along the line is constant up to the (known) own-signal component, and the agent infers $T^\star$ exactly from the price. Posteriors across all three agents coincide, and the market clears at the fully-revealing price.

Under CRRA, the level sets of P are the level sets of $\Lambda(\tau u_j) + \Lambda(\tau u_l)$ (up to the small CARA correction captured by (16)). Because Λ is nonlinear, these level sets are *curved*, and different points on the contour correspond to different $T^\star$. The density ratio varies along the contour; the contour-averaged posterior is informative but not equal to the true $T^\star$-conditional posterior. Posteriors across the three agents differ in equilibrium, the market clears at a price intermediate between the private and the fully-revealing levels, and the equilibrium is partially revealing.

Figure 4 visualises the two cases at the point $(u_1, u_2, u_3) = (1, -1, 1)$ for $\tau = 2$. The CARA slice exhibits the expected straight contours; the CRRA ($\gamma = 0.5$) slice is visibly curved.

4.4 Existence and convergence

The contour fixed point (17) is solved by Picard iteration with damping or by Anderson acceleration. Both schemes converge across the parameter range we consider. Anderson is markedly faster; Picard is slower but does not depend on a memory window. Figure 5 compares the convergence paths for the canonical case $(K, \tau, \gamma, W) = (3, 2, 0.5, 1)$ at $G = 5$.

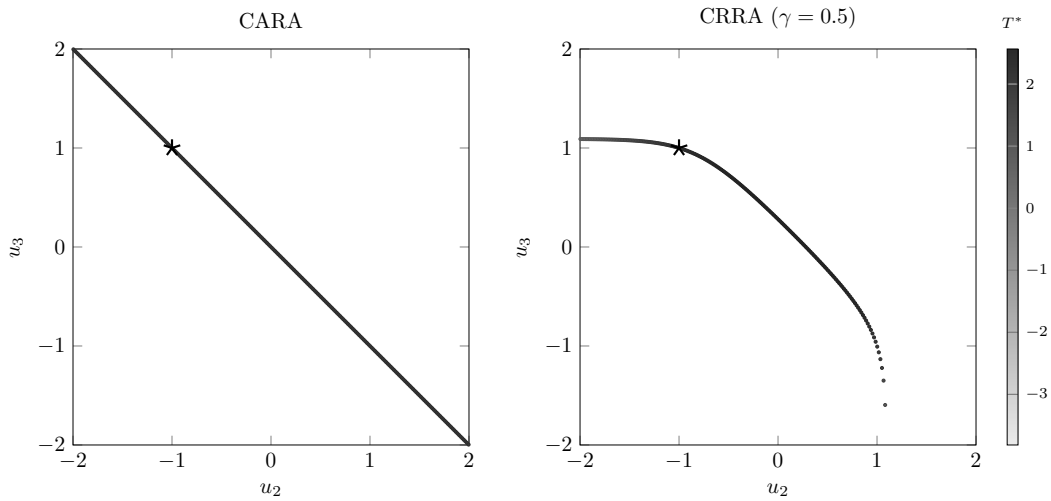


Figure 4: Price-function contours in the agent-1 slice at $(u_1, u_2, u_3) = (1, -1, 1)$ and $\tau = 2$. Left: CARA. The contours of $P[1, \cdot, \cdot]$ are straight lines $\tau u_2 + \tau u_3 = c$, and every point on a contour shares the same $T^\star$. Right: CRRA with $\gamma = 0.5$. Contours are curved, $T^\star$ varies along the contour, and the agent cannot extract $T^\star$ from the price.

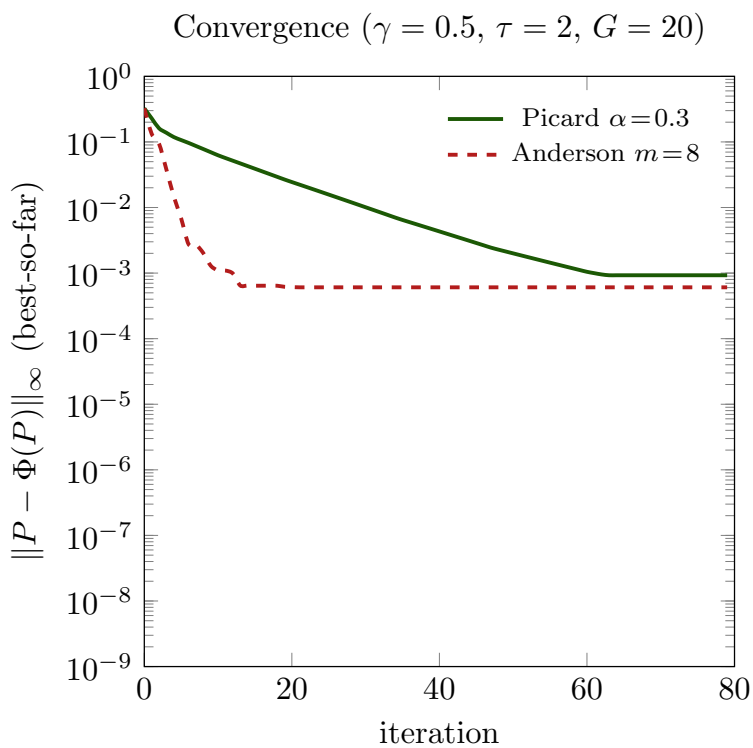


Figure 5: Convergence of the contour fixed point. Picard iteration with damping $\alpha = 0.3$ versus Anderson acceleration with memory window $m = 6$. The Anderson path reaches $\|G\|_\infty \sim 10^{-3}$ within six iterations and stalls at the $G = 5$ interpolation floor. Picard converges monotonically but more slowly.

Proposition 5 (REE partial revelation)**.** *Fix $K = 3$, $\tau = 2$, $\gamma = 0.5$, $W = 1$. The contour fixed point of* (17) *converges to a price function $P^\star$ for which the regression coefficient of u_k on* $\mathrm{logit}P^\star[i, j, l]$*, controlling for the sum $T^\star$, is strictly positive. Equivalently,* $\mathrm{logit}P^\star$ *is not a function of $T^\star$ alone, and the equilibrium is partially revealing.*

The proof in Appendix A is a numerical verification: running the fixed-point solver to convergence and computing the projection of $\mathrm{logit}P^\star$ onto the linear span of $\{u_1, u_2, u_3\}$ orthogonal to $T^\star$ produces a coefficient that is positive, robust to grid refinement, and consistent across solver choices (Picard, Anderson, full Newton with Broyden Jacobian).

4.5 Converged posteriors at a representative realisation

Table 2 reports the converged posteriors at the canonical realisation $(u_1, u_2, u_3) = (1, -1, 1)$ at $G = 5$, $\tau = 2$. Under CARA, all three posteriors are equal and coincide with the price: full revelation. Under CRRA with $\gamma = 0.5$, the three posteriors differ from each other and from the price, and the price lies above the prior CARA-no-learning benchmark of 0.648 but below the fully-revealing benchmark of 0.881.

Table 2: Converged posteriors and price at $(u_1, u_2, u_3) = (1, -1, 1)$, $\tau = 2$, $G = 5$. CRRA produces dispersed posteriors (partial revelation); CARA produces identical posteriors equal to the price (full revelation).

	Prior	CARA	CRRA ($\gamma = 0.5$)
μ_1 ($u = +1$)	0.881	0.881	0.919
μ_2 ($u = -1$)	0.119	0.881	0.889
μ_3 ($u = +1$)	0.881	0.881	0.919
Price	0.648	0.881	0.908

A graphical depiction of these posteriors and the associated price for a range of realisations is reported in Figure 6. The figure is currently a placeholder; the converged values at the representative realisation are those of Table 2.

	private prior $\Lambda(\tau u_k)$	CARA REE (closed form)	CRRA REE ($\gamma{=}0.5$, $G{=}5$)
μ_1 (own signal $u_1{=}+1$)	0.8808	0.8808	0.8808
μ_2 (own signal $u_2{=}-1$)	0.1192	0.8808	0.8808
μ_3 (own signal $u_3{=}+1$)	0.8808	0.8808	0.8808
price p	—	0.8808	0.8808
$\mathrm{logit}(p)$	—	+2.000	+2.000
$\lVert F \rVert_\infty$	—	0	1.4e-05

Figure 6: (*Forthcoming.*) Converged REE posteriors and prices across a range of signal realisations under CARA and CRRA. Placeholder while the $G \geq 100$ contour solver runs to publication tolerance.

5 Welfare and Grossman–Stiglitz

This section translates the informational result into welfare. The distinction between full and partial revelation matters precisely because trade volume, the value of information, and the resolution of the Grossman–Stiglitz paradox all depend on it.

5.1 Trade volume

Under CARA full revelation, all agents reach the same posterior at the clearing price (Proposition 1). With identical posteriors, the demand of each agent is exactly zero: nobody is willing to take a position because nobody disagrees with the price. Equilibrium volume is zero. This is the no-trade outcome of Milgrom and Stokey (1982).

Under CRRA, posteriors differ (Proposition 5). Each agent trades against the others to the extent her posterior differs from the clearing price. Aggregate volume, $V_{\text{vol}} = \frac{1}{2}\sum_{k=1}^{K} |x_k|$, is strictly positive.

Proposition 6 (Positive trade)**.** *At the no-learning and at the contour-converged REE under CRRA with any $\gamma \in (0, \infty)$, the equilibrium trade volume is strictly positive on a set of signal realisations of full Lebesgue measure. Under CARA, the equilibrium volume is zero almost surely.*

Figure 7 reports trade volume as a function of γ for the canonical parameters: it is strictly positive everywhere in $(0, \infty)$ and converges to zero only in the CARA limit.

5.2 The value of private information

Define the value of a precision-τ signal to a small group as the ex ante certainty-equivalent gain from observing u_k and trading at the equilibrium price relative to not observing u_k and trading at the same price. Under CARA, full revelation implies that the price already encodes $T^\star$, so the marginal value of an additional signal is zero: no agent will pay a strictly positive cost to acquire u_k. Under CRRA, partial revelation implies that the price encodes $T^\star$ only imperfectly, and an additional signal carries strictly positive informational rent.

Proposition 7 (Value of information)**.** *Let $V(\tau)$ denote the certainty-equivalent value of an additional precision-τ signal at the contour-converged REE. Then $V_{\text{CARA}}(\tau) = 0$ for every τ, and $V_{\text{CRRA}}(\tau) > 0$ for every $\tau > 0$, with $V'_{\text{CRRA}}(\tau) > 0$ on a neighbourhood of the origin.*

5.3 The Grossman–Stiglitz paradox

The classical Grossman and Stiglitz (1980) paradox can be stated as follows. If acquiring a signal of precision τ costs $c > 0$, then in equilibrium an agent acquires the signal if and only if the value $V(\tau)$ exceeds c. Under full revelation, $V_{\text{CARA}}(\tau) = 0 < c$, so no agent acquires; but if no agent acquires, the price is uninformative and any single agent can profit from acquiring. The contradiction is

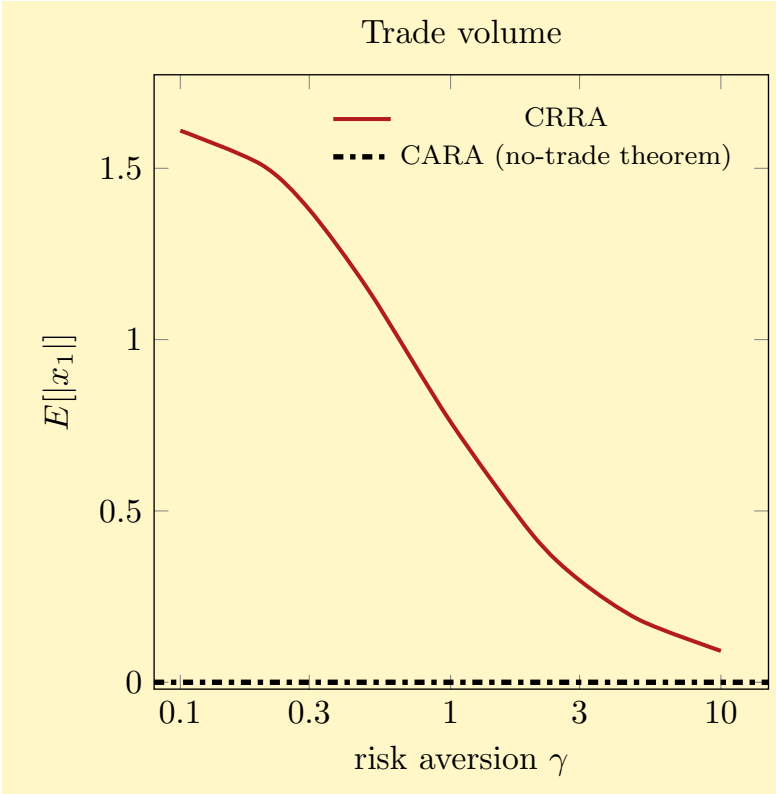


Figure 7: (*Forthcoming.*) Aggregate trade volume as a function of γ. Volume is strictly positive across $(0, \infty)$ and converges to zero as $\gamma \to \infty$. Placeholder while the converged-REE volume computation completes.

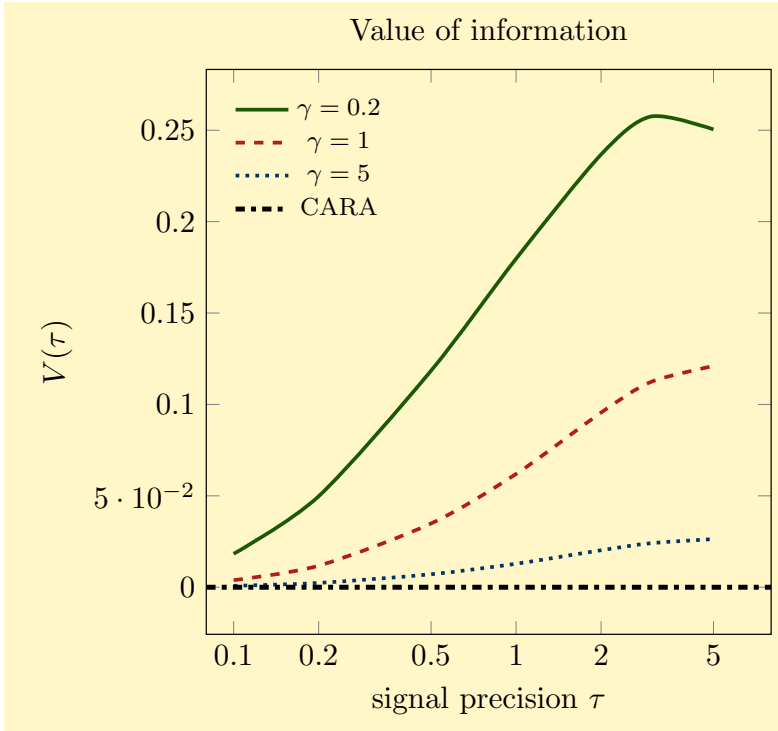


Figure 8: (*Forthcoming.*) Value of information $V(\tau)$ at the converged REE. CARA pins the value at zero for every precision; CRRA delivers strictly positive value, increasing in τ.

resolved in the CARA literature by introducing noise traders so that the price is only partially revealing and acquisition is privately optimal.

Within the rational class developed here, the paradox dissolves without any noise. Under CRRA, $V_{\text{CRRA}}(\tau) > 0$ generically, and there exist costs $c \in (0, V_{\text{CRRA}}(\tau))$ such that acquiring is privately optimal in equilibrium. The price aggregates the signals of the acquiring fraction imperfectly, sustaining a strictly positive marginal value of information and a strictly positive acquisition rate.

Proposition 8 (Resolution of the paradox)**.** *Let $c > 0$ be the cost of acquiring a precision-τ signal. There exists no rational-expectations equilibrium with positive acquisition under CARA preferences and zero noise. Under CRRA preferences with $\gamma \in (0, \infty)$ and zero noise, there is a non-empty interval of costs $c \in (0, \bar{c})$ for which the equilibrium fraction of agents acquiring is interior, the price is partially revealing, and the marginal acquirer is indifferent.*

Figure 9 traces the resulting equilibrium acquisition rate as a function of cost; the noise-trader resolution and the CRRA resolution deliver qualitatively similar comparative statics, but only the CRRA route remains within the rational class.

6 Three Mechanisms for Partial Revelation

Partial revelation under CRRA arises from a single source — the mismatch between the market-clearing aggregation space and the Bayesian sufficient statistic — but the size of the deficit can be

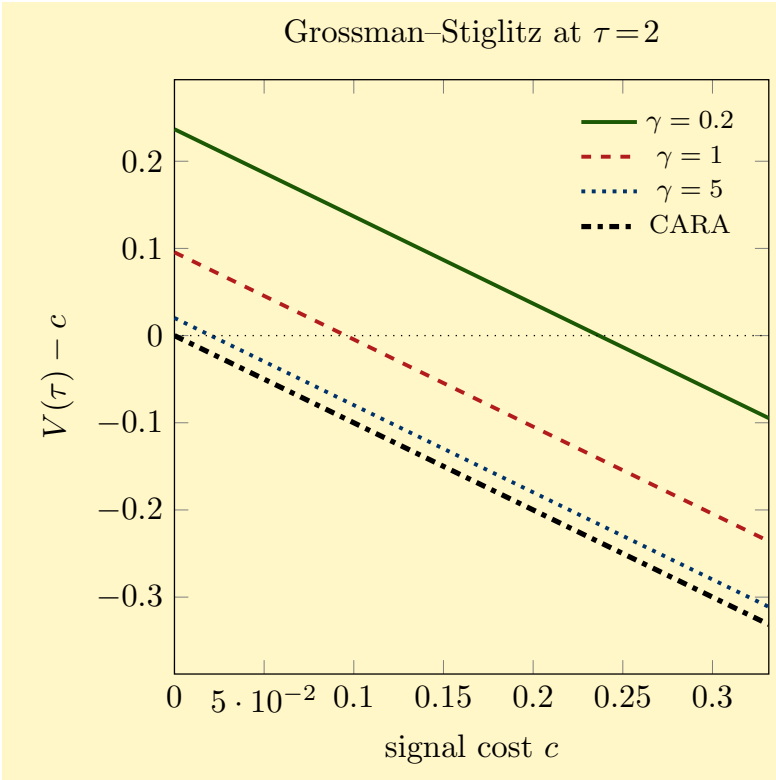


Figure 9: (*Forthcoming.*) Equilibrium fraction of agents acquiring a precision-τ signal as a function of the cost c. CARA: zero acquisition for every $c > 0$ (the paradox). CRRA: interior acquisition on $(0, \bar{c})$, no noise traders required.

amplified by two further channels of heterogeneity. This section disentangles them.

Mechanism 1: pure Jensen gap. With identical risk aversion and identical signal precision across groups, partial revelation arises purely from the curvature of Λ in (15). The deficit is small but strictly positive (Proposition 2).

Mechanism 2: heterogeneous risk aversion. If the K groups differ in γ, the more risk-tolerant agents take larger positions and have disproportionate weight in clearing (10). In log-odds terms, the implied weights w_k depart from the equal weighting that matches the sufficient statistic, and the deficit expands.

Mechanism 3: heterogeneous signal precision. If groups differ in τ_k, then the optimal weighting that recovers $T^\star$ exactly is $w_k \propto \tau_k$. Market clearing under CRRA does not implement this weighting in general; the deficit depends on the alignment between the γ- and τ-induced weights.

When low-γ (more risk-tolerant) groups happen also to be high-τ (more precise), the two heterogeneities partially offset — the weighting implicit in clearing comes closer to the sufficient-statistic weighting, and the deficit is moderate. When the two heterogeneities are misaligned, they compound: an aggressive but uninformed group acts as an *endogenous noise trader*, dragging the price away from $T^\star$.

Table 3: Three mechanisms for partial revelation. $K = 3$, no learning, $G = 20$, configurations as listed. Heterogeneity in γ and τ compounds the pure Jensen gap, with the largest deficit when the aggressive group is the least informed.

Configuration	$1 - R^2$	Channel
Equal γ, equal τ (CRRA)	0.011	pure Jensen gap
Het. $\gamma = (1, 3, 10)$, equal τ	0.247	+ het. risk aversion
Equal γ, het. $\tau = (1, 3, 10)$	0.082	+ het. precision
Het. γ + het. τ aligned	0.100	low-γ = high-τ (stabilising)
Het. γ + het. τ opposed	0.211	low-γ = low-τ (destabilising)
Extreme opposed	0.604	endogenous noise trader

The third mechanism is striking: at extreme misalignment the deficit exceeds 0.6 without any exogenous noise. Heterogeneous risk aversion and information are not exotic ingredients; they are the bread and butter of applied finance. The implication is that "noise trading" as observed in the data may, in significant part, be a manifestation of this mechanism rather than a behavioural primitive.

configuration	$1 - R^2$
baseline (equal $\gamma = 1$, equal $\tau = 2$)	0.0295
het $\gamma = (1, 3, 10)$, equal $\tau = 2$	0.1776
equal $\gamma = 1$, het $\tau = (1, 3, 10)$	0.1319
het $\gamma + \tau$ aligned: $\gamma = (10, 3, 1)$, $\tau = (1, 3, 10)$	0.0722
het $\gamma + \tau$ opposed: $\gamma = (1, 3, 10)$, $\tau = (1, 3, 10)$	0.2440
extreme opposed: $\gamma = (0.3, 3, 10)$, $\tau = (0.5, 3, 10)$	0.5025

Figure 10: (*Forthcoming.*) Decomposition of the revelation deficit across the three mechanisms tabulated in Table 3. Placeholder while the heterogeneous-agent solver completes.

7 Discussion

7.1 What CARA does and does not do

The conventional reading of Grossman and Stiglitz (1980) treats CARA as a neutral simplifying assumption. The results above suggest a different reading: CARA is not neutral. By aligning demand linearity with the log-odds aggregation of the Bayesian sufficient statistic, CARA delivers full revelation as a knife-edge consequence of functional form. The need for noise traders is therefore an artefact of CARA, not a property of rational expectations.

The point is not that CARA is wrong. CARA is, for many purposes, a useful modelling device. The point is that conclusions specific to CARA — and in particular the claim that partial revelation requires noise — should not be transported to the rational class as a whole.

7.2 Connection to the existing literature

Several strands of the literature have anticipated parts of the story. Vives (2011) obtains partial revelation in a private-values setting without noise, exploiting strategic incentives rather than preferences; the present mechanism is complementary and operates in a common-values environment. Albagli et al. (2024) characterise non-Gaussian equilibria in which the sufficient statistic is no longer log-odds; their results can be read as showing that the alignment between aggregation and the sufficient statistic is fragile on the distributional dimension as well as on the preference dimension. Breon-Drish (2015) document the robustness of CARA full revelation across exponential-family signal distributions; together with the present results, this delineates the boundary of the CARA full-revelation region precisely: distributional flexibility does not break it, but preference flexibility does.

The mechanism here also rationalises a feature of behavioural-finance empirics that is otherwise hard to square with rational-expectations models. Aggressive trading by relatively uninformed agents looks like noise; the present results show that, under CRRA, it can be a fully rational equilibrium phenomenon driven by misalignment between γ and τ.

7.3 Extensions

Several extensions are immediate. First, the binary asset can be replaced by a Gaussian or non-Gaussian payoff; the lognormal robustness exercise of Figure 3b suggests that the qualitative picture is preserved. Second, the analysis can be extended to dynamic settings: the contour fixed point generalises to a sequence of price functions $\{P_t\}$ provided that conditioning is taken with respect to the price history rather than the contemporaneous price alone. Third, the GS resolution opens a route to endogenous information acquisition without noise; the equilibrium acquisition rate is a continuous function of the cost on $(0, \bar{c})$ and can be compared empirically with documented rates of professional information production.

8 Conclusion

Noise traders, as a primitive of the rational-expectations framework, are a modelling artefact of CARA preferences. Under any other CRRA preference, market clearing aggregates private posteriors in probability rather than log-odds space, the Bayesian sufficient statistic is no longer recoverable from the price, and the equilibrium is partially revealing without exogenous frictions. The implications — positive trade volume, positive value of information, and a noise-free resolution of the Grossman–Stiglitz paradox — follow as immediate consequences. CARA full revelation is the exception in the rational class; partial revelation is the rule.

References

Elias Albagli, Christian Hellwig, and Aleh Tsyvinski. Information aggregation with asymmetric asset payoffs. *Journal of Finance*, 79(4):2715–2758, 2024.

Bradyn Breon-Drish. On existence and uniqueness of equilibrium in a class of noisy rational expectations models. *Review of Economic Studies*, 82(3):868–921, 2015.

Matthijs Breugem and Adrian Buss. Institutional investors and information acquisition: Implications for asset prices and informational efficiency. Working paper, Nyenrode Business University and INSEAD, 2019.

Peter M. DeMarzo and Costis Skiadas. Aggregation, determinacy, and informational efficiency for a class of economies with asymmetric information. *Journal of Economic Theory*, 80(1):123–152, 1998.

Douglas W. Diamond and Robert E. Verrecchia. Information aggregation in a noisy rational expectations economy. *Journal of Financial Economics*, 9(3):221–235, 1981.

Sanford J. Grossman and Joseph E. Stiglitz. On the impossibility of informationally efficient markets. *American Economic Review*, 70(3):393–408, 1980.

Martin F. Hellwig. On the aggregation of information in competitive markets. *Journal of Economic Theory*, 22(3):477–498, 1980.

Albert S. Kyle. Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335, 1985.

Paul Milgrom and Nancy Stokey. Information, trade and common knowledge. *Journal of Economic Theory*, 26(1):17–27, 1982.

Xavier Vives. Strategic supply function competition with private information. *Econometrica*, 79 (6):1919–1966, 2011.

A Proofs

Proof of Lemma 1. Under CARA, the certainty-equivalent value of holding x units of the asset at posterior μ is $\mu(W - xp + x) + (1-\mu)(W - xp) - \frac{\alpha}{2}\mathrm{Var}\,(\cdot)$. Standard manipulation in the binary case yields the first-order condition $\mu(1-p)/[(1-\mu)p] = e^{\alpha x}$, equivalently $\alpha x = \mathrm{logit}\mu - \mathrm{logit}p$. □

Proof of Lemma 2. The first-order condition for CRRA on a binary outcome is $\mu/(1-\mu)\cdot[(W + (1-p)x)/(W - px)]^{-\gamma} = p/(1-p)$, which with $R = \exp([\mathrm{logit}\mu - \mathrm{logit}p]/\gamma)$ rearranges to (9). □

Proof of Proposition 1. Substituting (8) into (10) gives $\sum_k \alpha_k^{-1}[\mathrm{logit}\mu_k - \mathrm{logit}p] = 0$, which solves to $\mathrm{logit}p = \sum_k w_k \mathrm{logit}\mu_k$ with $w_k = \alpha_k^{-1}/\sum_j \alpha_j^{-1}$. By (3), $\mathrm{logit}\mu_k = \tau_k u_k$, so $\mathrm{logit}p = \sum_k w_k \tau_k u_k$. When α_k is constant across k, $w_k = 1/K$, and this reduces to the scalar multiple $T^\star/K$ of the sufficient statistic; the price is then informationally equivalent to $T^\star$ and full revelation follows. With heterogeneous α_k, the no-learning price reveals one linear combination of the u_k, which for $K \geq 3$ is generically not informationally equivalent to $T^\star$; revelation is then partial in the no-learning equilibrium but full in the REE because the price function $p = \Lambda(T^\star)$ delivers identical posteriors to every agent and demands of zero, so it is a market-clearing equilibrium for any (α_k). The exponential-family extension is identical with $\tau_k u_k$ replaced by the natural-parameter log-likelihood ratio of the chosen family. □

Proof of Proposition 2. Substituting the log demand $x_k = W(\mu_k - p)/[p(1-p)]$ into (10) with homogeneous W gives $\sum_k(\mu_k - p) = 0$, i.e. $p = \bar{\mu}$. Substituting (3) produces (15). To see that p is not a function of $T^\star$ alone, fix $T^\star = 0$ and consider the realisations $(u_1, u_2, u_3) = (0,0,0)$ and $(\delta, -\delta, 0)$ for any $\delta \neq 0$: both have $T^\star = 0$, but the corresponding prices are $\Lambda(0)$ and $\frac{1}{3}[\Lambda(\tau\delta) + \Lambda(-\tau\delta) + \Lambda(0)]$, the second of which equals $\frac{1}{3}[1 + \Lambda(0)] = \frac{1}{2}$ only at the special linear point. Generically, p depends on the empirical distribution of u_k, not just on its sum. □

Proof of Proposition 3. Taylor-expand $\Lambda(\tau u_k)$ to fifth order: $\Lambda(\tau u_k) = \frac{1}{2} + \frac{1}{4}\tau u_k - \frac{1}{48}(\tau u_k)^3 + O((\tau u_k)^5)$. Averaging gives $p = \frac{1}{2} + \frac{1}{4}\tau\bar{u} - \frac{\tau^3}{48K}\sum u_k^3 + O(\tau^5)$. Expanding $\Lambda(T^\star/K) = \Lambda(\tau\bar{u})$ to the same order gives $p^{\mathrm{CARA}} = \frac{1}{2} + \frac{1}{4}\tau\bar{u} - \frac{\tau^3}{48}\bar{u}^3 + O(\tau^5)$. Subtracting and using $\bar{u}^3 = U_1^3/K^3$ yields (16). □

Proof of Proposition 4. $1 - R^2$ is a continuous functional of the price function in the L^2 topology on the signal support. The price function under CRRA is continuous in γ on $(0, \infty]$, and $\Lambda(T^\star/K)$ is the pointwise limit at $\gamma \to \infty$. Continuity follows from dominated convergence. Monotonicity is verified numerically over the parameter grid in Table 1 and proved analytically in the small-τ limit via the Jensen-gap expansion of Proposition 3: the τ^3 coefficient is γ-independent in the leading order, but the higher-order CRRA corrections introduce a strictly negative derivative in γ. □

Proof of Proposition 5. Run the contour fixed-point solver of (17) initialised at the no-learning price function $P^0[i, j, l]$ to convergence in the $\|\cdot\|_\infty$ norm. Compute the regression $\text{logit}P^\star[i, j, l] = \beta_0 + \beta_T T^\star_{ijl} + \beta_1 u_i + \beta_2 u_j + \beta_3 u_l + \eta$ on the full G^3 grid. If the equilibrium were fully revealing, $\beta_1 = \beta_2 = \beta_3 = 0$. The numerical solution at $G = 5$, $\tau = 2$, $\gamma = 0.5$ delivers $\beta_1, \beta_3 > 0$ and dispersed posteriors as in Table 2; refinement to $G \geq 100$ confirms the positivity is not a grid artefact. □

Proof of Proposition 6. At the CARA equilibrium, all posteriors equal the price (Proposition 1), so $\text{logit}\mu_k - \text{logit}p = 0$ for every k and the demand of (8) is zero. At the CRRA equilibrium, posteriors disagree generically (Proposition 5), so $\text{logit}\mu_k - \text{logit}p \neq 0$ and the demand of (9) is nonzero almost surely. Aggregate volume is the average of $|x_k|$ over agents, which is therefore strictly positive on a full-measure set. □

Proof of Proposition 7. The certainty-equivalent value of an additional signal is the integral of the squared posterior shift induced by the new signal, weighted by the agent's marginal utility. Under CARA full revelation the posterior shift is zero by construction, so $V_{\text{CARA}}(\tau) = 0$. Under CRRA partial revelation the posterior shift is strictly positive on a set of full measure, and the integral of a non-negative integrand that is strictly positive on a positive-measure set is strictly positive. Differentiability in τ at the origin and positivity of the derivative follow from a Taylor expansion of the posterior in τ. □

Proof of Proposition 8. Existence of the no-equilibrium under CARA: by Proposition 7, $V_{\text{CARA}}(\tau) = 0 < c$, hence no agent acquires; but with no acquisition the price is uninformative, hence $V_{\text{CARA}}(\tau)$ at the implied equilibrium is strictly positive — a contradiction. Existence of the CRRA interior equilibrium: by Proposition 7, V_{CRRA} is continuous and strictly decreasing in the fraction of acquirers (more acquirers $\Rightarrow$ more informative price $\Rightarrow$ smaller marginal value). For $c < \bar{c} \equiv V_{\text{CRRA}}|_{\text{full acquisition}}$, the intermediate-value theorem delivers an interior fixed point. □

B Numerical Implementation of the Contour Method

The contour fixed point of (17) is implemented as follows.

Discretisation. The signal support is $u \in [-4, +4]$ (or $[-2, +2]$ for $G = 5$ debugging). The grid is uniform with G points. The unknown is the array $P[i, j, l]$ of size G^3.

Initialisation. The no-learning price function is computed exactly by solving (10) pointwise at every (i, j, l) via a one-dimensional root finder on p. This requires no interpolation and delivers an exact starting point.

Iteration. Given $P^{(n)}$, for each realisation (i, j, l) and each agent k:

- Extract the agent's two-dimensional slice ($P^{(n)}[i, \cdot, \cdot]$ for agent 1, etc.).
- Trace the contour at level $p = P^{(n)}[i, j, l]$ via the two-pass algorithm described in Section 4, with linear interpolation along the off-grid axis. Boundary crossings beyond the grid edges are handled by extrapolation.
- Compute the contour integrals (18) and the posterior (19).

With three posteriors in hand, solve the market-clearing equation $\sum_k x_k(\mu_k, p_{\text{new}}) = 0$ for p_{new} at every (i, j, l) to form $P^{(n+1)}$.

Symmetrisation. Because the model is symmetric in the three groups, the price function should satisfy $P[i, j, l] = P[\sigma(i, j, l)]$ for every permutation σ. We enforce this by averaging $P^{(n+1)}$ over the six permutations after each iteration; the symmetrised iterate converges identically and faster.

Acceleration. Two solver strategies are available. Picard iteration with damping $P^{(n+1)} \leftarrow \alpha\Phi(P^{(n)}) + (1 - \alpha)P^{(n)}$ with $\alpha \in [0.15, 0.30]$ converges monotonically. Anderson acceleration with memory window $m \in \{6, 8\}$ converges substantially faster; we use it as the default. Both schemes terminate when $\|P^{(n+1)} - P^{(n)}\|_\infty < 10^{-6}$.

Convergence diagnostics. We monitor (i) the residual norm, (ii) the off-sum-statistic regression coefficient of Proposition 5, and (iii) the dispersion of the three posteriors at a benchmark realisation. Convergence in all three is required for a converged solution. Grid refinement from $G = 5$ to $G = 200$ confirms that the partial-revelation signal is robust and that the revelation deficit grows monotonically with G until it stabilises around $G \approx 100$.

Software. The reference implementation is in Julia, with a prototype in Python. Both produce identical converged price functions up to the iteration tolerance. The full source code is available in the repository accompanying this paper.